SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO5: *Create graphical models to analyze and infer patterns in complex probabilistic systems for structured decision-making.(BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:50

Code of Conduct:

I __________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Mock Interview

Q1. "Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?" (5 Marks – 8 Mins)

Q2. "Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?" (5 Marks – 8 Mins)

Q3. "If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?" (5 Marks – 8 Mins)

Q4. "During a technical interview, how would you explain the use of Conditional Random

Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)

Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)

Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)

Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)

Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)

Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)

Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts

Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Response to Questions	10	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriatel y

Mock Interview

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Date :- 29-08-2026

(1)
(A) **Bayesian Networks**

Bayesian Networks Predict Machine failures using the Probability of sensor values.

Example :- A factory machine has three sensors:

- Temperature = 95°c
- Vibration = High
- Noise = High.

The Baysein network combines these values and predicts 90%. Chance of Machine Failure

(2) **Markov Random Fields**

(A) Markov Random Fields remove noise by using information from neighbouring pixels.

Example :- A CCTV camera captures a noisy image at night. One pixel is black due to noise, but all surrounding pixels are grey. They model changes the black pixel to gray, making the image clearer.

(3) Hidden Markov Model

(A) A Hidden Markov Model recognize spoken words by analyzing the sequence of sounds.

Example :-

You say "Turn on the fan" The microphone records your voice, and the HMM converts the sound into the correct text:

"Turn on the fan."

(4) Conditional Random Fields.

(A) Conditional Random Fields label words correctly by considering the entire sentence.

Example :-

"Ravi works at Infosys."

- Ravi → person
- works → verb
- Infosys → organization.